# 4.7 Transformations of Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.7 (30-minute plan) · Student sheet

**I can…**

- describe transformations of polynomial functions.
- write rules for transformed polynomial functions.
- graph transformed polynomial functions.

**Vocabulary** · fill in the missing word

**Horizontal translation** — _____

**Vertical translation** — _____

**Reflections** — $-f(x)$ in the _____; $f(-x)$ in the y-axis

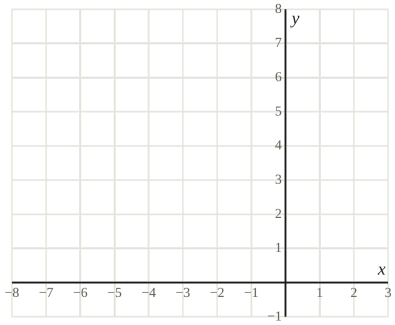
**Stretches and shrinks** — $a \cdot f(x)$ vertically; $f(ax)$ _____

## Part 1 · Notes and examples (fill in each note, then work the example)

**Translations:** $f(x - h)$ moves the graph _____; $f(x) + k$ moves it _____.

### Example 1

Describe the transformation of $f(x) = x^4$ represented by $g(x) = (x + 5)^4 + 1$.




**Reflections:** $f(-x)$ reflects in the _____; $-f(x)$ reflects in the _____.

### Example 2

Let $f(x) = x^3 - 6$. Write a rule for g, a vertical stretch by a factor of 2 followed by a translation 3 units up.

## Part 2 · Partner practice

### Partner practice

Describe the transformation of $f(x) = x^3$ represented by $g(x) = (x - 4)^3 - 7$.